

Describing the NHANES 2013 to 2018 Sample

BIOS 640: Assignment Week 4, Exercise 4

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Introduction

The National Health and Nutrition Examination Survey (NHANES) is run by the National Center for Health Statistics to assess the health and nutritional status of adults and children in the United States (Centers for Disease Control and Prevention and National Center for Health Statistics 2020). Each two year cycle, or wave, combines a household interview with a physical examination, where measurements such as blood pressure are taken.

This report describes 29,400 participants from the 3 waves covering 2013 to 2018. The data were cleaned in an earlier assignment, where examinations flagged as incomplete had their blood pressure readings set to missing.

NHANES uses some labels that are now considered outdated. The variable called gender records biological sex, and the ethnicity variable describes self-reported race. The original labels are kept here to match the source data.

Who is in the sample?

Table 1 shows how many people took part in each wave, along with the number and share of males and of each ethnic group. Percentages use every participant in the wave as the denominator.

Table 1: Sample size and demographic profile of NHANES participants by survey wave

Characteristic	Survey wave			All waves
	2013-2014	2015-2016	2017-2018	
Participants, n	10,175	9,971	9,254	29,400
Male, n (%)	5,003 (49.2%)	4,892 (49.1%)	4,557 (49.2%)	14,452 (49.2%)
Ethnicity, n (%)				
Mexican American	1,730 (17.0%)	1,921 (19.3%)	1,367 (14.8%)	5,018 (17.1%)
Other Hispanic	960 (9.4%)	1,308 (13.1%)	820 (8.9%)	3,088 (10.5%)
Non-Hispanic White	3,674 (36.1%)	3,066 (30.7%)	3,150 (34.0%)	9,890 (33.6%)
Non-Hispanic Black	2,267 (22.3%)	2,129 (21.4%)	2,115 (22.9%)	6,511 (22.1%)
Non-Hispanic Asian	1,074 (10.6%)	1,042 (10.5%)	1,168 (12.6%)	3,284 (11.2%)
Other race (incl. multiracial)	470 (4.6%)	505 (5.1%)	634 (6.9%)	1,609 (5.5%)

Note:

Percentages are calculated within each wave. Gender records biological sex, following the original NHANES label.

The three waves are similar in size, ranging from 9,254 to 10,175 participants. Males make up close to half of every wave, between 49.1% and 49.2%.

The ethnic make up is also stable from wave to wave. Non-Hispanic White participants form the largest group overall at 33.6%, and the remaining groups keep a similar share in each cycle. This consistency means differences between waves later on are unlikely to come from changes in who was sampled.

Systolic blood pressure

Average systolic blood pressure (SBP) is the mean of up to four readings per participant. People with no valid reading, mainly young children and those who were interviewed but never examined, are left out, so Table 2 is based on 21,604 participants.

Table 2: Summary statistics of average systolic blood pressure by survey wave

Wave	Participants	Average systolic blood pressure (mm Hg)				
		Minimum	Maximum	Median	Mean	SD
2013-2014	7,528	64.7	228.7	115.3	118.3	18.1
2015-2016	7,363	74.0	231.3	117.3	120.4	18.4
2017-2018	6,713	72.7	234.0	118.0	121.7	20.3
All waves	21,604	64.7	234.0	116.7	120.1	18.9

Note:

Participants are those with at least one valid SBP reading. SD is the standard deviation.

Mean average SBP stays within a narrow band across the three waves, from 118.3 mm Hg in 2013-2014 to 121.7 mm Hg in 2017-2018. The wide gap between the minimum and maximum in every wave reflects a sample that spans children through older adults.

Taken together, the waves are comparable in size, sex balance, ethnic make up and blood pressure. That makes it reasonable to pool them for analysis, as done in the “All waves” columns.

Tables were produced in R (R Core Team 2026) with R Markdown (Allaire et al. 2026), knitr (Xie 2025), dplyr (Wickham et al. 2026) and kableExtra (Zhu 2026).

References

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